

Unit 2

MACRONUTRIENTS

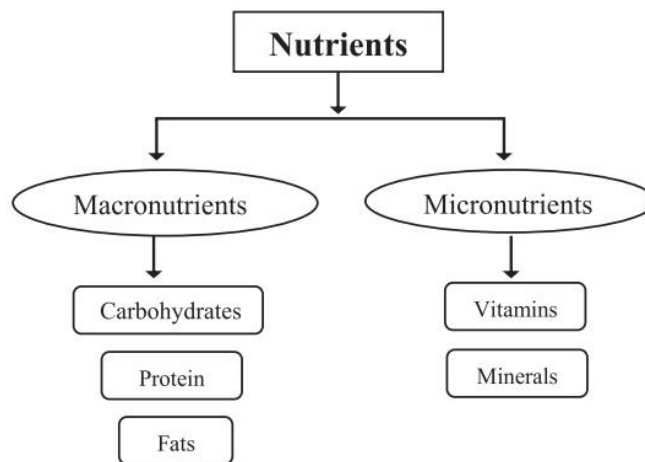
Classification, functions, Digestion & Absorption, RDA (Recommended Dietary Allowances), sources and deficiencies of Protein Carbohydrate and Fat, Dietary fiber - types and functions.

Introduction:

Nutrients

Nutrients are organic molecules that are needed by the body to grow, work, and stay healthy. All the cells of the body require these nutrients as fuels to function properly. Nutrients are digested from the food ingested and then broken down to be used by the organism.

The substances which are present in the food and consumed in our body for its vital functions are called nutrients. According to the World Health Organization (WHO), these nutrients must come from food, and they are vital for disease prevention, growth, and good health.



Classification of nutrients

Types of Nutrients

Nutrients are broadly classified into two types they are:

- **Macronutrients:** these are required in large amounts and are further classified into three types; carbohydrates, proteins, and fats.
- **Micronutrients:** these are required in small quantities and they consist of vitamins and minerals

Macronutrients

1. Proteins
2. Fats
3. Carbohydrates

Micronutrients

- Vitamins
- Microminerals

Macronutrients

Nutrients that are needed in large amounts are called **macronutrients**. There are three classes of macronutrients: carbohydrates, lipids, and proteins. Water is also a macronutrient in the sense that you require a large amount of it, but unlike the other macronutrients, it does not yield energy.

Macronutrients are nutrients needed in large amounts for energy, growth, and bodily functions, primarily classified into three main types: **Carbohydrates** (energy source, ~4 kcal/g), **Proteins** (building blocks, ~4 kcal/g), and **Fats/Lipids** (concentrated energy, ~9 kcal/g). While water is also needed in large amounts, it's not a calorie provider but crucial for bodily processes.

Classification:

Macronutrients include proteins, carbohydrates and fats.

1. Carbohydrates

Carbohydrates are molecules composed of carbon, hydrogen, and oxygen. The major food sources of carbohydrates are grains, dairy products, fruits, legumes, and starchy vegetables, like potatoes. Non-starchy vegetables, like carrots, also contain carbohydrates, but in lesser quantities.

Carbohydrates are broadly classified into two groups based on their chemical structure: simple carbohydrates (often called simple sugars) and complex carbohydrates, which include fiber, starch, and glycogen. Carbohydrates are a major fuel source for all cells of the body, and certain cells, like cells of the central nervous system and red blood cells, rely solely on carbohydrates for energy.

Carbohydrates are the chief sources of energy (Providing 4 Kcals/gm). Carbohydrates are cheap and readily available from food. There are three different kinds of carbohydrates. They include starch, sugar, and fiber. We do not get calories from fiber because our bodies do not break down fiber during digestion.

Classification of Carbohydrates

Monosaccharides: these are the simplest form of carbohydrates containing simple sugar molecule. **Example:** Glucose, Fructose and Galactose.

Disaccharides: These carbohydrates composed of two units of Monosaccharides. **Example:** Sucrose, Lactose and Maltose.

Polysaccharides: these are the complex sugars containing numerous units of monosaccharide molecules. **Example:** Glycogen, Cellulose and Pectin's.

Types of Carbohydrate

There are three types of carbohydrates:

1. Sugar
2. Starch
3. Fiber

1. Sugar

- Sugar is the simplest type of carbohydrate that provides energy rapidly.
- Our body quickly breaks down into glucose and absorbs simple carbohydrates, which results in the fluctuation of blood sugar levels.
- Glucose is the crucial energy source for our body's cells, tissues, and organs.
- Examples are white sugar, honey, glucose, agave nectar, cane syrup or corn syrup, molasses, etc.

There are two types of sugars:

a. Naturally occurring sugars: Milk and fresh fruits

b. Added sugars: Sweets (cookies, candy bars, ice cream), canned fruit, juice and soda

- According to the American Heart Association, people assigned female at birth (AFAB) should not consume more than 25 g (6 teaspoons or 100 calories) per day of added sugar.
- Similarly, people assigned male at birth (AMAB) should not consume more than 36 g (9 teaspoons or 150 calories) per day of added sugar.
- So, It is necessary to maintain the consumption of sugars due to the risk of causing diseases such as type 2 diabetes.
- As a result, it keeps your health in a good state.

2. Starch

- Starch (amylum), a natural component of plants, is a complex carbohydrate. It is a natural polymer consisting of a long chain of glucose molecules.
- The body breaks down the natural polymer into glucose- the energy source for the human body.
- There are two types: amylose and amylopectin.
- Amylose is a linear polymer, which is amorphous or solid. Amylopectin is a branched chain polymer, which is crystalline.
- It is present in many foods, such as grains, vegetables, and fruits (apples, berries, melons).
- Blood sugar remains stable because the body takes a long time to break down this complex carbohydrate.
- Examples are maize, tapioca, brown rice, oatmeal, roots, tubers, whole-wheat bread, pasta, corn, peas, wheat, potatoes, legumes, beans (black beans, chickpeas, lentils, lima, beans, and kidney beans), etc.

3. Fiber

- Fiber, a complex healthy carbohydrate or non-starch polysaccharide, does not break down glucose.
- Fiber can pass through the intestines and stimulate digestion. It is present in plant-based foods such as vegetables, fruits, and whole grains.
- According to the FDA (Food and Drug Administration), females should consume 28 grams (g) per day, whereas males should consume 34 grams.
- They are of two types: soluble and insoluble.
- Soluble fibers can dissolve in water, whereas insoluble one cannot. It helps to regulate blood sugar levels, feeds good bacteria in the gut, helps slow digestion, softens stool, and lowers cholesterol.
- Examples of high-fiber foods are Beans and legumes (black beans, chickpeas, lentils, lima beans, peanuts, and pinto beans), fruits (apples, seeds, and peaches), nuts and seeds (almonds, walnuts, pumpkin seeds, and sunflower seeds), whole-grain products (almonds, walnuts, pumpkin seeds, and sunflower seeds), and vegetables (corn, broccoli, brussels sprouts and squash).

Daily requirements

Carbohydrate intake should be in the range of 300-500gm (50%- 70%) out of the total energy intake for adults and 40-60% for children.

Sources: All sugars, jaggery, honey, pulses, whole grains, cereals, grains, rice, fruits, milk, yogurt, beans, roots and tubers such as potatoes, beet root etc.

Functions of Carbohydrates

- Supplies energy
- Protein sparing function
- Oxidation of fats
- Regulating blood glucose
- Facilitates bowel movements

Digestion and absorption of Carbohydrates

Salivary amylase aids digestion of starch in the mouth. Most of the digestion of carbohydrates takes place in the small intestine. Carbohydrates are absorbed into the blood stream as glucose, Galactose and fructose. By way of the capillaries of the villi, the simple sugars enter the portal circulation and transported to the liver.

Disorder of excess intake of carbohydrate

1. It can cause obesity.
2. It can cause dental caries.
3. It can increase insulin levels that may cause an increased tendency for blood clotting.

Proteins

Proteins are large molecules composed of chains of amino acids, which are simple subunits made of carbon, oxygen, hydrogen, and nitrogen. Food sources of proteins include meats, dairy products, seafood, and a variety of plant-based foods, like beans, nuts, and seeds. The word protein comes from a Greek word meaning “of primary importance,” which is an apt description of these macronutrients as they are also known as the “workhorses” of life. Proteins provide structure to bones, muscles, and skin, and they play a role in conducting most of the chemical reactions occurring in the body. Scientists estimate that more than 100,000 different proteins exist within the human body. Proteins can also provide energy, though this is a relatively minor function, as carbohydrates and fat are preferred energy sources.

Proteins are the indispensable constituents of the diet. Proteins are made up of amino acids. Amino acids are needed for replacement and growth of the body parts.

Amino acids are classified as essential and non-essential amino acids. Essential amino acids cannot be synthesized by the body and must be taken through foods whereas non-essential amino acids can be synthesized by the body.

Types of Protein

There are two types of protein. They are Complete protein and Incomplete protein.

a. Complete protein:

- Complete proteins are proteins that contain essential amino acids.
- Plant-based foods do not require complete protein.
- Examples are animal foods such as meat, fish, milk, and eggs.

b. Incomplete protein:

- Incomplete proteins are proteins that contain all the essential and non-essential amino acids.
- Examples are vegetable foods such as cereals, legumes, and peanuts.

Daily requirements

The ICMR recommends 1gm of protein/ Kg of body weight for adults.

The amount of protein should be increased for children, pregnant and lactating mothers by 1.5- 2 g/kg body weight.

Sources of Protein

Animal sources: eggs, milk and milk products, fish, meat.

Plant sources: pulses, cereals, dry fruits, beans nuts etc.

Functions of Proteins

- Protein helps in synthesis of enzymes, immunoglobulin, plasma proteins and hormones in the body
- Protein helps in growth and repair of body tissues
- Proteins are secondary sources of energy during deficiency of carbohydrates and fats. (Provides 4 Kcals/gm)
- Proteins help in forming haemoglobin
- Proteins help in antibody formation

Digestion and absorption: The hydrolysis of proteins in the gastro intestinal tract is accomplished by proteases secreted in gastric juice and pancreatic juice and also by proteases present in the intestinal mucosa.

Fats

Fat is deposited as adipose tissue in the body and perform essential functions in the body. Fats are composed of fatty acids and contain oxygen, carbon and hydrogen.

Types of fats

There are three types of fat. They are:

1. Saturated fats
2. Unsaturated fats
3. Trans fats

1. Saturated fats

- Saturated fats (solid at room temperature) are non-essential fatty acids, including butyric acid, palmitoleic and stearic acids.
- It is a nutrient from all animal foods, such as animal fats, coconut, and palm oils.
- But ingesting too much fat is harmful to our health.
- According to the American Heart Association, saturated fat consumption should not be less than 7 percent of calories.
- If people consume high fat, then the cholesterol level of a person increases in arteries.

- Examples are meat products (sausage, bacon, beef, hamburgers), pizza, dairy products (cheese, whole and reduced-fat milk, butter, and dairy desserts), cookies, grain-based desserts, oil of coconut, palm and kernel, and mixed fast food dishes.

2. Unsaturated fats

- Unsaturated fats, liquid at room temperature, are essential and non-essential fatty acids, which include linoleic acid, α -linolenic acid, EPA and DHA, GLA, and oleic acid.
- It is a nutrient from all vegetable oils except coconut and palm oil.
- It decreases LDL cholesterol levels and stabilizes heart rhythms.

Types of unsaturated fats

There are two types of unsaturated fats. They are:

1. Monounsaturated fats

- It consists of omega-9 fatty acids.
- Examples are oils (olive oil, peanut oil, canola oil), Avacados, nuts (almonds, hazelnuts, pecans), and seeds (pumpkin, sesame).

1. Polyunsaturated fats

- It consists of omega-3 fatty acids and omega-6 fatty acids.
- The three types of omega-3 fatty acids are as follows:
 - eicosapentaenoic acid (fish, shellfish, seafood)
 - docosahexaenoic acid (fish, shellfish, seafood)
 - α -linolenic acid (flax seed, soybean, walnut, rapeseed oil)
- Examples are oils (safflower, sunflower, corn, soy), walnuts, flax seeds, fish, and canola oils.
- Omega-3 fatty acids are innocuous to our health because they help lower triglyceride levels, raise HDL (good) cholesterol, lower blood pressure, and decrease the risk of premature death.

3. Trans fats

- Trans fats, unhealthy fats, are solid at room temperature.
- The source of trans fats in diets is partially hydrogenated oil. It is satisfying for taste; however, it harms the health of our bodies.
- It can raise LDL cholesterol levels and lower HDL (good) cholesterol levels in the blood.
- In addition, Experts should use trans fat in limited amounts in packaged foods and restaurants.
- Examples are processed foods (margarine spreads, peanut butter, chips, crackers, cookies, non-dairy creamer, pre-prepared cake frostings, vegetable shortening, candy bars), pie crusts, pizza dough, cookie dough, pastries, donuts, pies, fried and fast foods (pastries, breakfast foods, biscuits), naturally occurring trans fats (beef, lamb, butter, milk, cheese, yogurt).

Daily Requirements

15-20% of total calorie requirements should be from fat

Sources

Animal sources: Fish, egg, meat, milk and milk products.

Plant sources: oil seeds (Groundnut, mustard, cotton seed and coconut oil), nuts.

Functions

- Supplies energy (9 kcals/ gm)
- Improve the palatability of food (flavor and taste)
- Supports body organs like liver and kidneys
- Provides insulation and thermoregulation against cold
- Provides essential fatty acids which helps in growth, promotion and maintenance of skin integrity
- Helps in formation of hormones in the body
- Helps in transportation of fat soluble vitamins

Digestion and absorption

There is no digestion of fats in mouth and very little in stomach. In the small intestine, presence of fat stimulates cholecystokinin and secretin which further stimulates pancreatic juice and bile. Food mixes with bile and emulsified. Pancreatic lipase hydrolyses and yields fatty acids and monoglycerides. These pass to small intestine and combined with intestinal lumen and absorbed in to blood stream.

Excess intake of fats

1. It can cause obesity.
2. It can cause atherosclerosis and CHD due to high LDL levels and high triglycerides in the blood.
3. It can cause diabetes mellitus type II and gallbladder stones.

Water

There is one other nutrient that we must have in large quantities: **water**. Water does not contain carbon but is composed of two hydrogens and one oxygen per molecule of water. More than 60 percent of your total body weight is water. Without it, nothing could be transported in or out of the body, chemical reactions would not occur, organs would not be cushioned, and body temperature would fluctuate widely. On average, an adult consumes just over two liters of water

per day from food and drink combined. Since water is so critical for life's basic processes, we can only survive a few days without it, making it one of the most vital nutrients

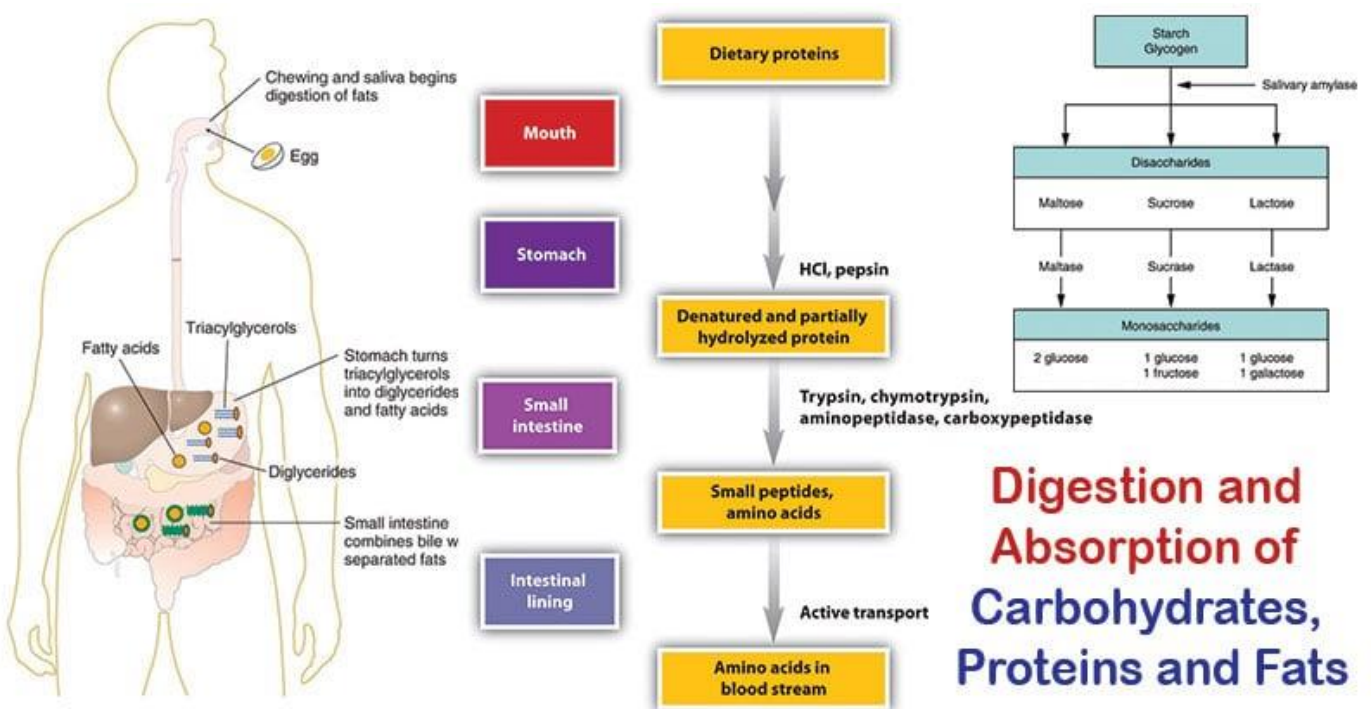
People should consume 2.7 to 3.7 liters/day of water.

Functions of Water

It is necessary for our health and has many functions, such as:

1. It helps in digestion, absorption, circulation, and excretion processes.
2. It acts as a solvent for body constituents.
3. It regulates body temperature.
4. It helps in the lubrication of the moving body parts.
5. It carries nutrients.

Digestion & Absorption of Protein Carbohydrate and Fat



Digestion breaks food into absorbable units: carbohydrates into simple sugars (monosaccharides), proteins into amino acids/small peptides, and fats into fatty acids/monoglycerides, mainly in the small intestine with help from bile and pancreatic enzymes. Absorption moves these nutrients into the bloodstream (sugars, amino acids via portal vein) or lymph (fats as chylomicrons) for energy, building, and storage.

Carbohydrate Digestion & Absorption

- **Mouth:** Salivary amylase starts breaking down starches into smaller sugars (maltose).
- **Stomach:** Acid deactivates amylase.
- **Small Intestine:** Pancreatic amylase continues breakdown; intestinal enzymes finish turning them into glucose, fructose, galactose.
- **Absorption:** Monosaccharides are actively or passively absorbed into the blood and go to the liver.

Protein Digestion & Absorption

- **Mouth:** No protein digestion.
- **Stomach:** Pepsin (activated by HCl) starts breaking proteins into smaller polypeptides.
- **Small Intestine:** Pancreatic proteases (like trypsin) break polypeptides into amino acids, dipeptides, and tripeptides.
- **Absorption:** These are absorbed by intestinal cells, further broken into amino acids, and enter the blood.

Fat (Lipid) Digestion & Absorption

- **Mouth/Stomach:** Lingual and gastric lipases begin some fat breakdown (triglycerides).
- **Small Intestine (Main Site):** Bile salts emulsify fats, and pancreatic lipase breaks them into free fatty acids and monoglycerides.
- **Absorption:** Fatty acids and monoglycerides reform into triglycerides within intestinal cells, combine with protein and phospholipids to form chylomicrons, which enter the lymph system, then the blood.

Deficiencies of Protein Carbohydrate and Fat

Deficiencies in protein, carbs, and fats lead to severe malnutrition (Marasmus), causing extreme weight loss, muscle wasting, fatigue, and stunted growth, while isolated deficiencies cause issues like skin problems (fat), hair/nail issues (protein), and low blood sugar/energy (carbs). Protein lack affects muscle/brain; fat lack impacts skin/cognition; carb lack reduces immediate energy, causing fatigue, dizziness, and headaches.

Protein Deficiency

- **Symptoms:** Muscle wasting, weakness, brittle hair, dry skin, edema (swelling), impaired growth, mood changes (depression/aggression).
- **Diseases:** Kwashiorkor (protein-specific) and Marasmus (combined protein/calorie deficiency).
- **Impact:** Body can't build neurotransmitters, collagen, or keratin; causes fatigue and anemia.

Carbohydrate Deficiency

- **Symptoms:** Fatigue, weakness, dizziness, headaches, low blood sugar (hypoglycemia).
- **Impact:** Lack of primary fuel for the body, leading to it breaking down protein and fat for energy.

Fat Deficiency (Essential Fatty Acids)

- **Symptoms:** Dry, scaly skin (dermatitis), slow wound healing, potential cognitive issues (learning deficits).
- **Impact:** Essential for brain, eye health, and skin integrity.

Combined Deficiencies (Marasmus)

- A severe lack of all macronutrients (protein, carbs, fats) results in extreme emaciation, severe underweight, sunken face, lack of body fat, and stunted development.

Marasmus

Marasmus is a severe form of malnutrition — specifically, protein-energy undernutrition. It results from an overall lack of calories. Marasmus is a deficiency of all macronutrients: carbohydrates, fats, and protein. If you have marasmus, you lack the fuel necessary to maintain normal body functions. People with marasmus are visibly depleted, severely underweight and emaciated. Children may be stunted in size and development. Prolonged marasmus leads to starvation.

What is the difference between marasmus and kwashiorkor?

Marasmus and kwashiorkor are two different variations of severe protein-energy undernutrition. Marasmus is a deficiency of all macronutrients, while kwashiorkor is a deficiency in protein

predominantly. Kwashiorkor occurs in people who may have access to carbohydrates — bread, grains or starches — but lack protein in their diet. Marasmus has a wasted and shriveled appearance, while kwashiorkor is known for causing edema — swelling with fluid, especially in the belly and the face.

Marasmus affect:

Marasmus can affect anyone who lacks overall nutrition, but it particularly affects children, especially infants, who require more calories to support their growing bodies. It is more common in developing countries with widespread poverty and food scarcity, and where parasites and infectious diseases may contribute to calorie depletion. In the developed world, elderly people in nursing homes and hospitals or who live alone with few resources are more at risk.

What happens to the body in marasmus disease?

When the body is deprived of energy from food, it begins to feed on its own tissues — first adipose tissue (body fat) and then muscle. It also begins shutting down some of its functions to conserve energy. Cardiac activity slows down, causing low heart rate, low blood pressure and low body temperature. In some cases, this leads to heart failure. The immune system is also compromised, making undernourished people more prone to infection and illness and slower to recover.

Children with chronic marasmus will not have the physical resources to grow and develop as they should. They may be stunted in size or have developmental delays or intellectual disabilities. These effects can be lasting, even in children who receive treatment. Parts of the digestive system also begin to atrophy from the lack of use. This means that even when people do have food to eat, they might not be able to absorb nutrition from their food effectively. Ironically, marasmus can lead to food aversion.

Symptoms and Causes

The main causes affecting all ages include:

- Poverty and food scarcity.
- Wasting diseases such as AIDS.
- Infections that cause chronic diarrhea.
- Anorexia.

Additional causes affecting children include:

- Inadequate breastfeeding or early weaning of infants.
- Child abuse/neglect.

Additional causes affecting adults include:

- Dementia.
- Elder abuse/neglect.

External signs of marasmus

- Visible wasting of fat and muscle.
- Prominent skeleton.
- Head appears large for the body.
- Face may appear old and wizened.
- Dry, loose skin (skin atrophy).
- Dry, brittle hair or hair loss.
- Sunken fontanelles in infants.
- Lethargy, apathy and weakness.
- Weight loss of more than 40%.
- BMI below 16.

Management and Treatment

Stage 1: Rehydration and stabilization

The first stage of treatment is focused on treating dehydration, electrolyte imbalances and micronutrient deficiencies to prepare the body for refeeding. In many cases, these can all be treated with one formula, REhydration SOLution for MALnutrition (ReSoMal), given orally or through a nasogastric tube. It's also important to keep the person warm to prevent hypothermia and to treat infections, which compromise their meager energy resources. Depending on the individual, it may take several hours to days before they are considered stable enough to begin refeeding.

Stage 2: Nutritional rehabilitation

Refeeding begins slowly with liquid formulas that carefully balance carbohydrates, proteins and fats. For inpatients, healthcare providers prefer tube feeding because it allows for gradual but continuous nutrition. Calories are introduced at about 70% of normal recommended values for the person's age. Eventually, they may increase to 140% of recommended values to meet the growth requirements of stunted children. This phase may last two to six weeks. During this time, patients gradually progress to more ordinary oral feeding with solid foods.

Stage 3: Follow-up and prevention

Since marasmus can recur, a complete treatment protocol includes education and outgoing support for the patient and/or their caregiver before they are discharged. In the developing world, this may mean breastfeeding support, safe drinking water and food preparation guidelines, immunizations and education to prevent widespread diseases. In the developed world, caregivers may need guidance on how to recognize signs of malnutrition in those they care for. The Malnutrition Universal Screening Tool (MUST) can help identify people at risk.

RDA (Recommended Dietary Allowances)

Recommended Dietary Allowances (RDAs) are the levels of intake of essential nutrients that, on the basis of scientific knowledge, are judged by the Food and Nutrition Board to be adequate to meet the known nutrient needs of practically all healthy persons.

Protein, carbs, and fats provide energy and support body functions, with RDAs varying by age/activity (e.g., protein ~0.8g/kg body weight, carbs ~130g minimum for brain). Sources include meat, dairy, legumes (protein); grains, fruits, veggies (carbs); oils, nuts, fatty fish (fats). Deficiencies cause issues like muscle loss (protein), fatigue (carbs), or fatty acid problems (fat), while excess can lead to obesity, so balance with whole foods, fiber, and healthy fats (unsaturated) is key.

Protein

- **RDA:** Around 0.8 g/kg body weight for adults, but higher for active individuals/children; it's crucial for tissues.
- **Sources:** Meat, poultry, fish, eggs, dairy, beans, lentils, tofu, nuts, seeds.
- **Deficiency:** Muscle wasting, weakened immunity, edema (kwashiorkor), stunted growth in children.

Carbohydrates

- **RDA:** Minimum 130g/day for brain function, but higher needs exist; WHO recommends coming mostly from whole foods.
- **Sources:** Whole grains (oats, brown rice), fruits, vegetables, legumes, starchy roots.
- **Deficiency:** Fatigue, lack of energy, impaired brain function.
- **Excess:** Weight gain, increased risk of diabetes/heart disease.

Fat

- **RDA:** No strict RDA, but AMDR is 20-35% of calories; aim for minimal saturated/trans fats, focus on unsaturated.
- **Sources:** Avocado, nuts, seeds, olive oil, fatty fish (omega-3s), dairy, meat (saturated).
- **Deficiency:** Hormonal issues, poor absorption of fat-soluble vitamins (A, D, E, K).
- **Excess:** Obesity, heart disease (especially from saturated/trans fats).

Recommended Dietary Allowances and RDA Of Indians

In a family there may be infants, young children, adolescent, adults, and elderly all living under one roof. It is always a big challenge to provide a good nutritious diet for each of the member, particularly when their needs, preferences vary. So what should be the guiding factor to ensure balanced meals that meet the needs of all members? Is there a Standard or a Reference that would serve as a goal for Good Nutrition? This section focuses on this important concept of Recommended Dietary Allowances.

Basic concept of recommended daily allowance:

Humans need a wide range of nutrients to lead a healthy and active life. The amount of each nutrient needed for an individual depends on age, body weight, physical activity, physiological state (pregnancy, lactation) etc.

The requirement for a particular nutrient is the minimum amount that needs to be consumed to prevent symptoms of deficiency and to maintain satisfactory level of the nutrient in the body.

For example in case of infants and children, the requirement may be equated with the amount that will maintain a satisfactory rate of growth and development. Similarly for an adult the nutrient requirement is the amount that will maintain body weight and prevent the depletion of the nutrient from the body which otherwise may lead to deficiency. In physiological condition like pregnancy and lactation, adult women may need additional nutrients to meet the demand of fetal growth along with their own nutrient needs.

Now within each group (say infants or an adults etc) there may be individual variations in the nutrient requirements. For instance, there may be a period of low intake or the quality of the diet may vary, similarly the effect of cooking and processing may be different and bioavailability of the nutrient from the diet may also vary.

REQUIREMENT + SAFETY MARGIN = RECOMMENDED DIETARY INTAKE

The Recommended Dietary Allowances (RDA) are the levels of intake of the essential nutrients that are judged to be adequate or sufficient to meet the nutrient requirement of nearly all (97 to 98 percent) healthy individuals in a particular life stage and gender group

From our discussion above it must be clear to you that the Nutrient Requirement of an individual and the Dietary Allowance for a group or a population are distinctly different. RDA takes into account the variability that exists in the requirement of a given nutrient between individuals in a given population group. So RDA is neither minimal requirement nor necessarily optimal level of intake. Rather, RDA is the safe and adequate level, which incorporates margin of safety intended to be sufficiently generous (high enough) to encompass the presumed variability in requirements among individuals and meet the needs of almost all healthy people.

Further please note, RDA's do not apply to people who are suffering from disease which influence the nutrient intake. They only apply to healthy people.

Significance / Uses of RDA

RDA, we know, represents the level of the nutrient to be consumed daily to meet all the requirements of most of the individuals in a given population. So RDA's help us plan balanced diets which include a variety of foods derived from diverse food groups which help meet the nutrient requirements. Other than this basic use, RDA's have come to serve many important purposes. The various applications of RDA include:

- Comparison of individual intakes to the RDA allows an estimate to be made about the probable risk of deficiency among individuals,
- Modifying nutrient requirements in clinical management of diseases,
- To help public health nutritionists to compose diets for schools, hospitals, prisons etc.
- For health care policy makers and public health nutritionists to design, develop nutrition intervention programmes and policies,
- For planning and procuring food supplies for population groups,
- For evaluating the adequacy of food supplies in meeting national nutritional needs
- For interpreting food consumption records of individuals and populations,
- For establishing Standards for the national feeding programmes implemented by the Governments for its vulnerable population,
- For designing nutrition education programmes for the masses,
- For developing new food products and dietary supplements by the industry,

- Establishing guidelines for the national labeling of packaged foods (by Food Standards Safety Authority of India (FSSAI))

Recommended Dietary Allowances (RDA) for Indian Population:

For the Indian population, the dietary standards have been computed by the Indian Council of Medical Research (ICMR). These recommendations have been published as "Nutrient Requirements and Recommended Dietary Allowances for Indians" (ICMR 2010)

The recommendations are constantly revised whenever new data is available. The last recommendations were revised in 2010, based on the new guidelines of the International Joint FAO/WHO/UNU Consultative Group and based on the data on Indians that had accumulated after 1989 recommendations. To help you understand this recommendation here are a few highlights:

1. Note, the RDA for Indians are presented for the different age categories: 0-6 months, 7 to 12 months, 1 – 3 years, 4 – 6 years, 7 – 9 years, 10 – 12 years, 13 – 15 years, 16 – 18 years, adult man and women.
2. Recommendations are given for energy and all other nutrients including proteins, visible fat, calcium, iron, retinol, Beta Carotene, thiamine, riboflavin etc.
3. Recommended dietary allowances for adults are based on sex (male, female), body weight and physical activity level (i.e. Sedentary, Moderate and Heavy work) .
4. RDA for energy is expressed in kilocalories (Kcal), for proteins, fats in grams (g), and for calcium, iron, vitamins and minerals in milligram (mg) or microgram.
5. RDA for protein is based on body weight. The relationship can be expressed as 1g protein per kg body weight in the case of adults. It varies for other age categories.
6. RDA for energy and protein are given as additional intakes in pregnancy and lactation, indicated by a (" + " sign). This requirement is over and above the normal requirement of adult women. RDA for other nutrients are given as total intake figures.
7. In infancy RDA's for energy, protein, iron, thiamin, riboflavin and niacin are expressed as per kg body weight (expected for a healthy, normal growing infant of a particular age)
8. RDA for Vitamin A have been given in terms of retinol or alternatively in terms of Beta Carotene.

Table 1a. Summary of recommended energy requirement for Indians

Age group	Category	ICMR 2020	ICMR 2010	Difference
		Kcal/d		
Adult Men	Sedentary work	2110	2320	-210
	Moderate work	2710	2730	-20
	Heavy work	3470	3490	-20
Adult Women	Sedentary work	1660	1900	-240
	Moderate work	2130	2230	-100
	Heavy work	2720	2850	-130
	Pregnant	+ 350	+ 350	--
	Lactating (0-6m)	+600	+600	--
	Lactating (7-12m)	+520	+520	--
Infants	0-6 months	530	520	+10
	6-12 months	660	670	-10
Children*	1-3 y	1070	1060	+10
	4-6 y	1360	1350	+10
	7-9 y	1700	1690	+10
Boys	10-12 y	2220	2190	+30
Girls	10-12 y	2060	2010	+50
Boys	13-15 y	2860	2750	+110
Girls	13-15 y	2400	2330	+70
Boys	16-18 y	3320	3020	+300
Girls	16-18 y	2500	2440	+60

*For children 1% increase over the previous recommendations.

Table 2a. Summary of recommended protein requirements for Indians 2020

		Body weight (kg)	EAR (g/kg/d)	RDA (g/kg/d)	EAR (g/d)	RDA (g/d)	
Adult Men	Sedentary	65	0.66	0.83	42.9	54.0	
	Moderate						
	Heavy Work						
Adult Women	Sedentary	55	0.66	0.83	36.3	45.7	
	Moderate						
	Heavy Work						
Pregnant Women	2nd Trimester				+7.6	+9.5	
	3rd Trimester				+17.6	+22.0	
Lactating Women	0-6 months				+13.6	+16.9	
	6-12 months				+10.6	+13.2	
Infants	0-6 months	5.8	1.16	1.40	6.7	8.1	
	6-12 months	8.5	1.04	1.23	8.8	10.5	
Children	1-3y	12.9	0.79	0.97	10.2	12.5	
	4-6y	18.3	0.70	0.87	12.8	15.9	
	7-9y	25.3	0.75	0.92	19.0	23.3	
Boys	10-12y	34.9	0.75	0.91	26.2	31.8	
Girls	10-12y	36.4	0.73	0.90	26.6	32.8	
Boys	13-15y	50.5	0.72	0.89	36.4	44.9	
Girls	13-15y	49.6	0.70	0.87	34.7	43.2	
Boys	16-18y	64.4	0.70	0.86	45.1	55.4	
Girls	16-18y	55.7	0.67	0.83	37.3	46.2	

For people consuming cereal-based diet with low quality protein, the protein requirements are 1 g/kg per day

Note: The cereal-legume-milk composition of the diet should be 3:1:2.5 for good protein quality.

Dietary fiber - types and functions.

Fiber is found mainly in plant foods such as fruits, vegetables, whole grains and members of the bean family called legumes. Fiber may be best known for its ability to prevent or relieve constipation. But foods with fiber can have other good effects as well. They can help you stay at a healthy weight and lower the risk of diabetes, heart disease and some types of cancer.

Choosing tasty foods that provide fiber isn't hard. Find out how much dietary fiber you need, which foods have it, and how to add them to meals and snacks.

What is dietary fiber?

Dietary fiber is a nutrient known as a carbohydrate. Fiber includes the parts of plant foods that the body can't digest or absorb. This makes it different from nutrients such as fats, proteins, and other carbohydrates including starches and sugars. The body breaks down these nutrients and absorbs them. Instead, fiber passes somewhat intact through the stomach, small intestine and colon and out of the body.

There are two main types of fiber:

- **Soluble fiber.** This type of fiber dissolves in water. It forms a gel-like material in the stomach that slows down digestion. It can help lower cholesterol and blood sugar. Soluble fiber is found in oats, peas, beans, apples, bananas, avocados, citrus fruits, carrots, barley and psyllium.
- **Insoluble fiber.** This type of fiber doesn't dissolve in water. It supports the movement of material through the digestive system and adds bulk to stool. So it can be helpful for people who have constipation or don't regularly pass stool. Some good sources of insoluble fiber are whole-wheat flour, wheat bran, nuts, beans, and vegetables such as cauliflower, green beans and potatoes.

Most high-fiber plant foods contain both soluble and insoluble fiber. The amount of each type of fiber varies with the type of plant, such as whether it's a fruit, vegetable or whole grain. But you can get both types of fiber by eating a variety of fiber-rich foods.

Key Functions & Benefits

- **Digestive Health:** Promotes regular bowel movements, prevents constipation, and reduces risk of hemorrhoids and diverticulitis.
- **Blood Sugar Control:** Slows carbohydrate absorption, preventing spikes in blood sugar.
- **Heart Health:** Helps lower "bad" LDL cholesterol by binding to bile acids.
- **Weight Management:** Increases feelings of fullness (satiety), helping with appetite control and weight loss.

- **Gut Microbiome:** Fermentable fibers feed beneficial gut bacteria, supporting a healthy gut.
- **Cancer Prevention:** Dilutes potential carcinogens and speeds their passage, reducing colon cancer risk.

Recommended Intake

- About 25 grams per day for women and 38 grams for men (or 14g per 1000 calories).

The National Academy of Medicine gives the following daily fiber recommendations for adults:

- 21 grams for women older than age 50.
- 25 grams for women age 50 or younger.
- 30 grams for men older than age 50.
- 38 grams for men age 50 or younger.

Benefits of a high-fiber diet

A high-fiber diet may support good health in a number of ways.

High fiber and less chance of constipation

Dietary fiber increases the weight and size of stool and softens it. In general, bulky stool is easier to pass, and this lowers the chance of constipation. If you have loose, watery stools, fiber may help make them solid. That's because fiber absorbs water and adds bulk to stool.

Bowel health and fiber

A high-fiber diet may lower the risk of swollen veins in the anus and lower rectum called hemorrhoids. Eating plenty of fiber also may help lower the risk of a condition called diverticulitis that involves small, inflamed pouches in the colon wall. A high-fiber diet is linked with a lower risk of colorectal cancer as well. Some fiber also serves as food for "good" bacteria in the gut. It's known as fermented fiber. It may play a role in lowering the risk of diseases of the colon.

High fiber and lower cholesterol levels

Soluble fiber found in beans, oats, flaxseed and oat bran may keep the body from absorbing some of the cholesterol in other foods. As a result, that may lower low-density lipoprotein, also called "bad," cholesterol levels in the blood. High-fiber foods may have other effects on heart health, such as lowering blood pressure and lessening swelling in the body called inflammation.

Blood sugar levels and fiber

In people with diabetes, fiber may slow the absorption of sugar and help improve blood sugar levels. That's especially true of soluble fiber. A healthy diet that includes both soluble and insoluble fiber also may lower the risk of type 2 diabetes.

Healthy weight and fiber

High-fiber foods tend to be more filling than low-fiber foods. So you're likely to eat less and stay satisfied longer. High-fiber foods also tend to take longer to eat and to be less energy dense than low-fiber foods. That means they have fewer calories for the same volume of food.

High-fiber diet and lifespan

Getting more fiber is linked with a lower risk of dying of any health condition, including heart disease.

Best fiber food:

Nutritious foods that can help you get more daily fiber include:

- Whole-grains such as barley, bulgur, brown rice, and whole-wheat bread or pasta.
- Fruits.
- Vegetables.
- Beans, peas and other legumes.
- Nuts and seeds.

Refined or processed foods usually are lower in fiber. The grain-refining process removes the outer coat, called bran, from the grain. That lowers the amount of fiber and other nutrients in the grain.

Examples of refined grains include white bread, pasta and cereals that aren't made with whole grains. Also, removing the skin from fruit or the pulp from fruit juice lowers fiber. Enriched foods have some of the B vitamins and iron added back after processing, but not the fiber.

Fiber supplements and fortified foods

In general, whole foods are better for you than fiber supplements. Fiber supplements don't provide the variety of fiber, vitamins, minerals and other healthful nutrients that whole foods do.

Your healthcare professional may recommend a fiber supplement if diet changes don't add enough fiber to your diet or if you have certain health conditions. These include conditions that affect the intestines such as constipation, diarrhea and irritable bowel syndrome. Check with your healthcare professional before taking fiber supplements. These supplements include psyllium (Metamucil, Konsyl, others), methylcellulose (Citrucel) and calcium polycarbophil (FiberCon).

Another way to get more fiber is to eat food products that have fiber added to them by foodmakers. Some added fiber comes from plant sources such as bran, husks or roots that have been processed. Other types of added fiber are made in a lab. Added fiber ingredients that you

might see on food labels include chicory root, cellulose and pectin. Some people say they get gas after eating foods with added fiber.

Tips for fitting in more fiber

Need ideas for adding more fiber to your meals and snacks? Try these suggestions:

- **Jump-start your day.** For breakfast, choose a breakfast cereal with 5 grams or more of fiber a serving. Opt for cereals with whole grain, bran or fiber in the name. Or add a few tablespoons of wheat bran that hasn't been processed to your favorite cereal. Top your cereal with sliced bananas, berries or other fruit.
- **Switch to whole grains.** Each day, make sure that at least half of the grains you eat are whole grains. Choose breads with at least 2 grams of fiber per serving. On the label, whole wheat, whole-wheat flour or another whole grain should be first in the ingredient list. Try other whole grains, too, such as brown rice, wild rice, buckwheat, whole-wheat pasta and quinoa.
- **Bulk up baked goods.** Substitute whole-grain flour for half or all of the white flour when baking. Try adding crushed bran cereal, wheat bran or uncooked oatmeal to muffins, cakes and cookies.
- **Learn to like legumes.** Beans and members of the bean family such as peas and lentils are excellent sources of fiber. Add kidney beans to canned soup or a green salad. Or make nachos with black beans, lots of fresh veggies, whole-wheat tortilla chips and salsa.
- **Eat fruit and vegetables.** They're rich in fiber as well as vitamins and minerals. Try to eat five or more servings daily. You can have fresh, frozen or canned fruits and vegetables. If you eat canned fruits, choose those that are canned in 100% fruit juice instead of syrup. If you eat canned vegetables, make sure they're low in sodium.
- **Make snacks count.** Whole fruits, raw vegetables, low-fat popcorn without salt and whole-grain crackers are all healthy choices. A handful of nuts or dried fruits that don't have added sugar also can be a healthy, high-fiber snack. Just be aware that crackers, nuts and dried fruits are high in calories.

High-fiber foods are good for your health. But adding too much fiber too quickly can lead to gas, bloating and cramping. Add fiber to your diet slowly over a few weeks. This lets natural bacteria in your digestive system adjust to the change. Also, drink plenty of water. Fiber works best when it absorbs water. This makes your stool soft, bulky and easier to pass.
